Data-driven Estimates of Structural Image Quality

Adon F. G. Rosen[a], David R. Roalf[a], Kosha Ruparel[a], Jason Blake[a], Kevin Seelaus[a], Lakshmi P. Villa[a], Rastko Ciric[a], Phillip A. Cook[a], Christos Davatzikos[a], Mark A. Elliot[a], Angel Garcia de La Garza[a], Eftathios D. Gennatas[a], Megan Quarmley[a], J. Eric Schmitt[a], Russell T. Shinohara[a], M. Dylan Tisdall[a], R. Cameron Craddock[b], Raquel E. Gur[a], Ruben C. Gur[a] & Theodore D. Satterthwaite[a]

[a] Perelman School of Medicine, University of Pennsylvania, Philadelphia PA, USA

[b] Center for the Developing Brain, Child Mind Institute, New York NY, USA

Background

Subject motion has been shown to be a major confound in many neuroimaging studies. Moreover, motion co-varies with variables of interest such as age. A recent study established that motion as measured by volumetric navigators directly impacts structural imaging measures. However, most studies to date have not been acquired with such navigator technology, and quantitative measures of structural image quality are lacking. The goal of this study was to identify quantitative measures of structural image quality that could be derived from the T1 volume alone when motion estimates were not present, and evaluate the confounding influence of data quality on inferences regarding brain development.

Methods

Figure 1: Training protocol for manual raters

- 1840 T1 structural images manually rated on 3 point scale.
- Quantitative metrics were produced with QAP and FreeSurfer.
- ROC curves probed classification ability of automated quality measures predicting usability derived from manual ratings.
- Univariate relationships were then explored between imaging outcomes and continuous estimates of image quality

Study	N	% Female	Age Mean	Age SD
Training	1065	51	14.90	3.70
Internal Testing	533	44	15.10	3.68
External Testing	242	48	41.36	16.99

Table 1. Participant demographics

Quantitative Metric	Abbreviation	Citation
Signal-to-noise ratio	SNR	Magnotta and Friedman, 2006
Contrast-to-noise ratio	CNR	Magnotta and Friedman, 2006
Foreground-to-background energy ratio	FBER	NA
Quality index 1	Qi1	Mortamet et al., 2009
Image smoothness	FWHM	Friedman et al., 2006
Entropy focus criterion	EFC	Atkinson etal., 1997
Euler number	Euler	Dale et al., 1999

Table 2. Quantitative metrics

Results

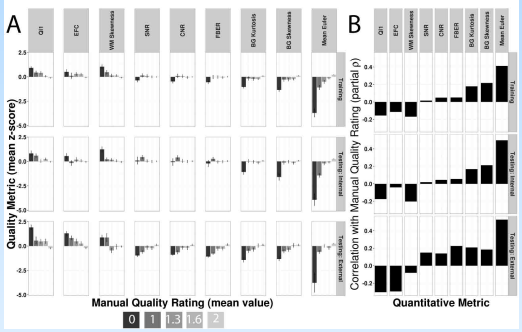

Figure 2. Quantitative metrics of image quality show heterogeneous alignment with mean manual rating.

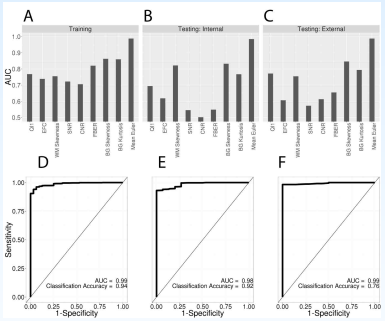

Figure 3. Inclusion model to identify unusable images. In all datasets, the Euler number was the best-performing metric; adding additional metrics to the Euler number did not improve model performance.

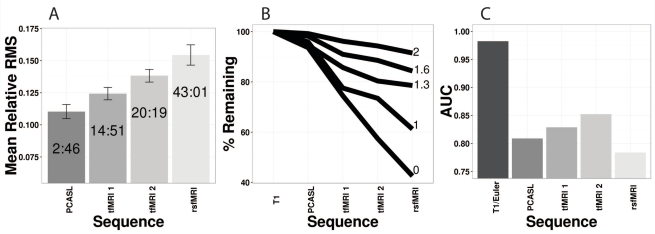

Figure 4. Limits of motion from functional scans as a proxy measure of T1 volume quality. Motion estimates increase with time, and subject attrition increases as quality decreases.

Results

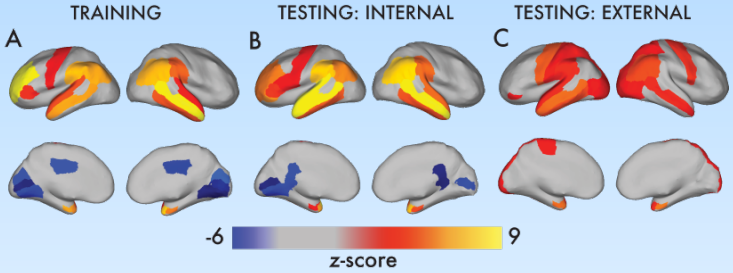

Figure 5. Quantitative measure of image quality is associated with cortical thickness. In usable images that were not excluded due to gross artifact, cortical thickness was significantly related to the Euler number in a regionally heterogeneous pattern.

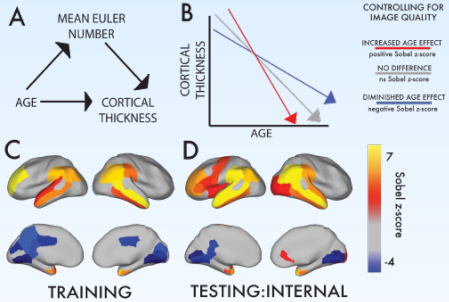

Figure 6. Data quality significantly mediated the relationship between age and cortical thickness in a bidirectional, regionally heterogeneous manner.

Discussion

Moving forward, these results suggest that data quality can be directly measured from structural images. Specifically, the Euler number can be use to successfully discriminate useable and unusable images. After removing unusable images, quantitative estimates of data quality can be used as a covariate in group-level analyses of structural imaging data. Future work should explore the relationship between these continuous estimates of image quality in relation to psychopathology.

Support & Disclosures

This work was supported by grants from the National Institute of Mental Health: R01MH107703 (TDS), R01MH112847 (TDS & RTS), R01MH107235 (RCG), R01MH112070 (CD), K01MH102609 (DRR), R01NS085211 (RTS), K01ES026840 (JES). Additional support was provided by the Dowshen Program for Neuroscience and the Penn/CHOP Lifespan Brain Institute. The PNC was funded through NIMH RC2 grants MH089983 and MH089924 (REG). Support for developing statistical analyses (RTS & TDS) was provided by a seed grant by the Center for Biomedical Computing and Image Analysis (CBICA) at Penn.

bioRxiv link

@roseadnn

Satterthwaite Lab